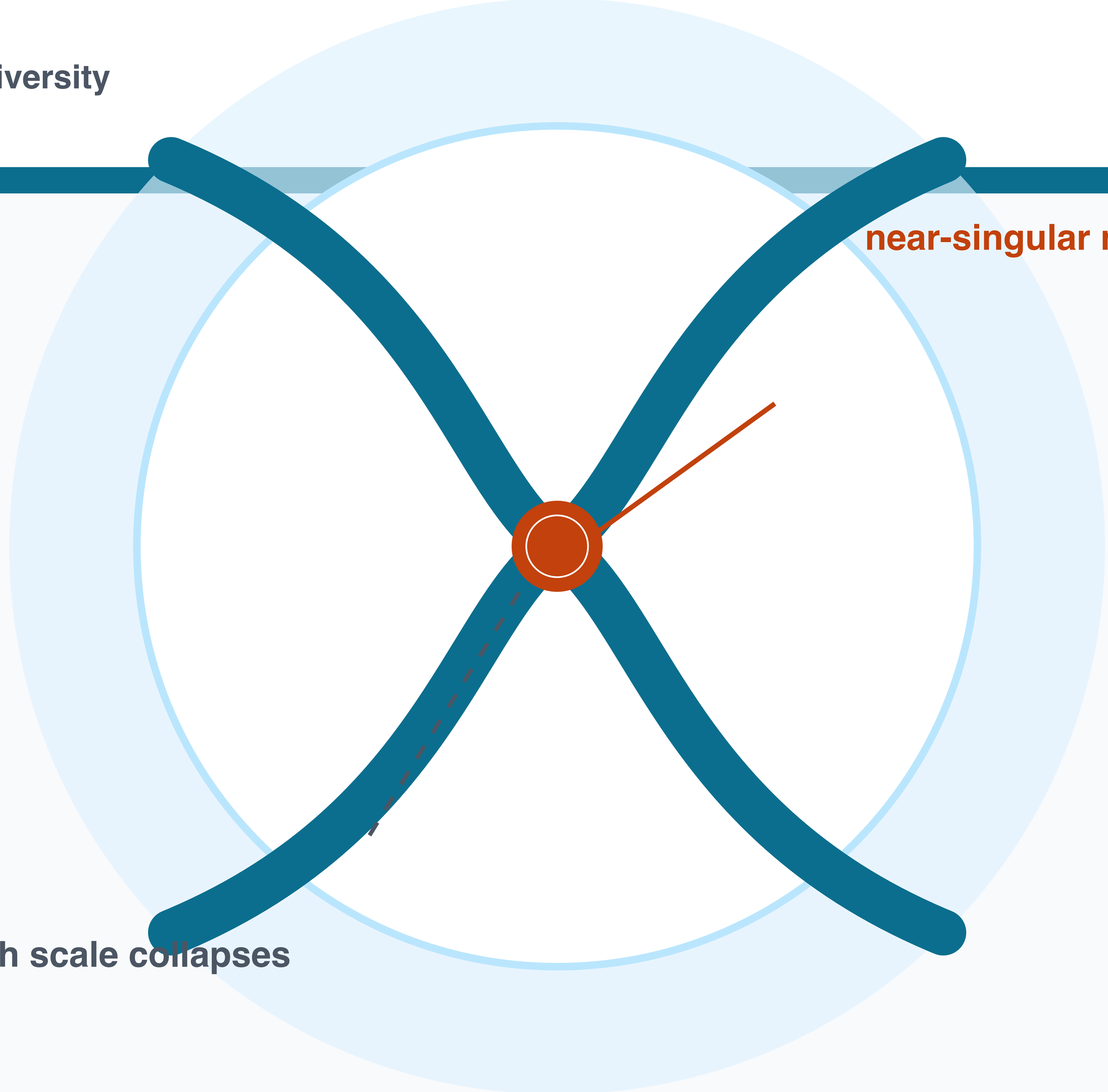
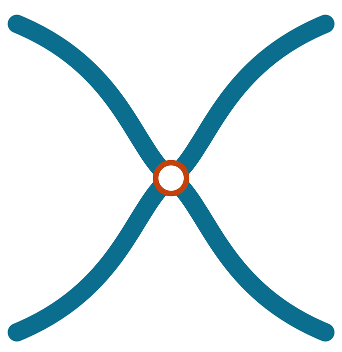
Hydrodynamic Singularities

When smooth flows focus geometry, stress, and time into a tiny region

CoMPhy Lab | Department of Physics | Durham University

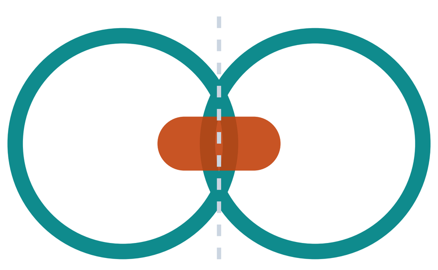
A0 scaffold | replace placeholders
with final figures and narrative





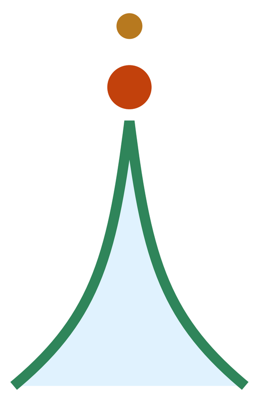
Pinch-off

minimum radius selects the clock



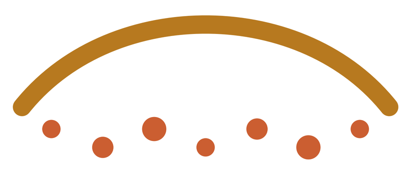
Coalescence

a microscopic bridge reshapes both drops



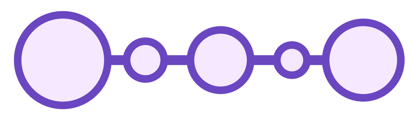
Jets

focusing launches fast tips and droplets



Sheets

rims, holes, and ligaments compete



Elastic threads

rheology changes the route to breakup

1. What becomes singular?

A smooth interface develops a neck, bridge, rim, or tip whose local length scale becomes much smaller than the surrounding flow. The global object looks simple; the decisive physics lives in a small inner region.

2. What resolves it?

Inertia, viscosity, surface tension, gas flow, contact-line physics, elasticity, and molecular cutoffs can each set the final balance. The useful question is not just whether breakup happens, but which balance controls the path.

3. Why should students care?

A tiny region controls printing, spraying, foams, emulsions, bubbles, jets, and soft biological flows. Singularities are where continuum mechanics shows both its power and its limits.

Scaling breadcrumbs for later content

Capillary time

t_c ~ sqrt(rho R^3 / sigma)

Pinch-off

r_min -> 0 in a self-similar inner flow

Coalescence

bridge radius couples local curvature to global motion

Soft matter

Oh, We, Ca, De, Wi decide the dominant balance

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